Unit 5, Lesson 5: Applying the Triangle Inequality Theorem

Benchmark

MA.8.GR.1.3 Use the Triangle Inequality Theorem to determine if a triangle can be formed from a given set of sides. Use the converse of the Pythagorean Theorem to determine if a right triangle can be formed from a given set of sides.

Learning Targets

- ☐ I can use the Triangle Inequality Theorem to determine if a given set of values makes a triangle.
- ☐ I can use the Triangle Inequality Theorem to determine possible values for the third side of a triangle when given two sides.

5.5.1 Warm-Up: Would You Rather?

The dimensions of two slices of pizza cut into triangles are given in the table.

	Slice A:	Slice B:
Height	2.8 inches (in.)	2.5 in.
Base	6.2 in.	6.6 in.

1. Explain which slice of pizza you would rather eat.

2. Which slice of pizza has the larger area?

5.5.2 Collaboration: Exploring the Triangle Inequality Theorem

1. Put both of your elbows on your desk. Face your hands toward each other, then move them down until they are flat on the desk with only the fingertips touching. Slowly raise your hands up without moving your elbows.

   a. What do you notice about the length of your arms and the distance between your elbows?

   b. What do you wonder about creating a triangle with these lengths?

2. Roll a die to pick 3 random numbers. Record your numbers in the blanks below. Then, circle true or false to indicate if the inequality is true or not.

   a. ______ (Roll 1) + ______ (Roll 2) > ______ (Roll 3) true or false

   b. ______ (Roll 2) + ______ (Roll 3) > ______ (Roll 1) true or false

   c. ______ (Roll 1) + ______ (Roll 3) > ______ (Roll 2) true or false

   d. Did you have any false answers?

3. Find a partner who has a different answer for Part D. Discuss what you notice about how different values affected the truth of the inequalities. Summarize your discussion.

4. Roll a die to pick 2 random numbers. Record your numbers on the blanks labeled for each roll. Leave any other lines blank.

a. ______ + ______ > ______
Roll 1 / Roll 2

b. ______ + ______ > ______
Roll 2 / Roll 1

c. ______ + ______ > ______
Roll 1 / Roll 2

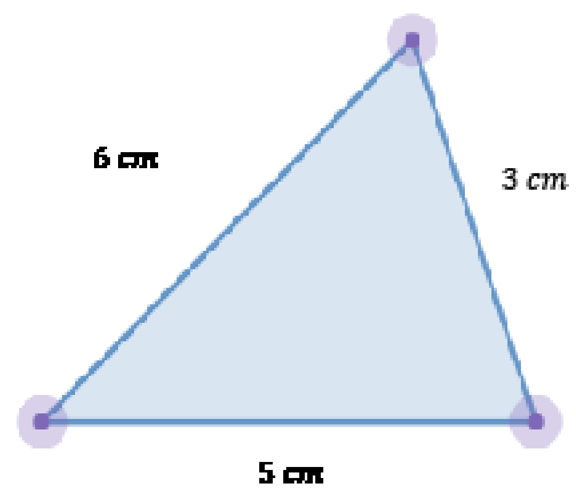
d. Determine one value that will satisfy all three inequalities based on the values already inserted from rolling the die.

5. Complete the statement.

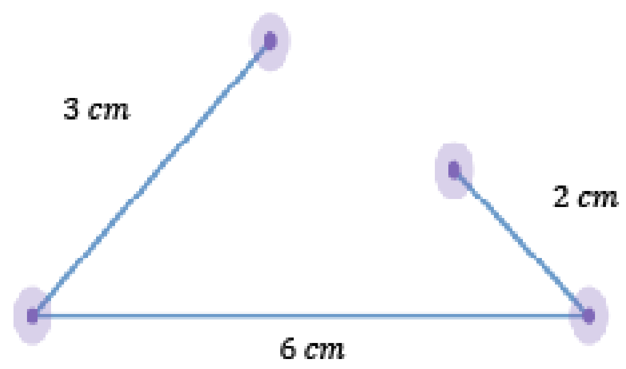
The [○ Triangle Inequality Theorem / ○ Pythagorean Theorem] says any [○ side / ○ angle] of a triangle is [○ more / ○ less] than the [○ difference / ○ sum] of the other two sides.

If any side is greater than or equal to the sum of the other 2 sides, a triangle will not be formed.

An example and a non-example are shown.



$3 + 6 > 5$	✓
$3 + 5 > 6$	✓
$6 + 5 > 3$	✓



$2 + 3 > 6$	X
$2 + 6 > 3$	✓
$3 + 6 > 2$	✓

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5.5.3 Guided Practice: Applying the Triangle Inequality Theorem

1. Use the given sets of side lengths in the table to determine if a triangle can be formed. Use the explanation column to explain or show why or why not.

Side Lengths	Can the side lengths form a triangle?	Explanation
10, 2, 9	o Yes o No	
12, 2, 16	o Yes o No	
6, 9, 12	o Yes o No	
7, 6, 13	o Yes o No	

2. Two side lengths of a triangle are given. Determine a third side length that could be used to form a triangle.

   a. 3, 4, ____

   b. 1, 5, ____

   c. 10, 20, ____

   d. 300, 400, ____

   e. Find a partner who has different answers than you on any problem. Discuss whether both your answers are correct. Write a summary of your discussion.

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5.5.4 Your Turn!

1. Determine if the following sets of side lengths can create a triangle.

   a. 3 in., 9 in., and 8 in.

   b. 16 ft., 6 ft., and 2 ft.

   c. 7 yd., 5 yd., and 10 yd.

2. Adriana is creating a picture of a house with popsicle sticks. She wants to glue a roof in the shape of a triangle. Circle the set of sides that could form her triangular roof.

   3 in., 5 in., 8 in.

   4 in., 4 in., 9 in.

   3 in., 3 in., 6 in.

   5 in., 6 in., 7 in.

3. Determine a third side that will form a triangle with the given side lengths.

   8, 5, ______

4. Maria is installing a garden fence in the shape of a triangle. She already has two pieces of fencing measuring 3 feet (ft.) and 4 ft. What is a possible length she can use for her third side?

5.5.5 Wrap-Up: Error Analysis

Darion was asked to determine if the following side lengths would form a triangle: 4 ft., 6 ft., and 10 ft. Darion said that they would form a triangle because $4 + 6$ is not less than 10.

1. Help Darion understand and learn from his mistake. Write a brief note to Darion explaining what his mistake is and how he could learn from it.

